

SE LAB 3

UML COMPONENT DIAGRAM

NAME : POOJITHA LAKSHMI BHASKARA RAGHVENDRA

SRN : PES1UG24AM384

DATE : 4TH September 2026

Pharmacy Expiry & Re-order Dispatch Engine

Architecture Selection

Architectural Choice

The Layered Architecture is suitable for the pharmacy system because the system contains clearly separated responsibilities such as the user interface, pharmacy operations, inventory and expiry management, and database storage. Each layer can focus on a specific part of the system while interacting with the layer directly below it.

Reason 1 – Clear Separation of Responsibilities

The architecture separates different pharmacy operations into distinct layers. The Pharmacy Staff User Interface handles user interactions, while components such as Medicine Batch & Inventory Management, FEFO Medicine Dispensing, Expiry Monitoring & Alerts, and Automated Re-order Management handle the corresponding business operations. The Pharmacy Database is responsible for storing and retrieving system data. This makes the system easier to understand and maintain.

Reason 2 – Easier Maintenance and Modification

The separation between layers allows individual functionality to be modified without significantly affecting other parts of the system. For example, the FEFO dispensing logic or expiry-alert mechanism can be updated while keeping the user interface and database structure largely independent. This makes future changes to the pharmacy system easier to implement.

Security Advantage

Layered Architecture provides a clear boundary between the user interface and the database. Pharmacy staff interact with the system through the application layer rather than directly accessing stored inventory and batch information. This helps restrict direct access to sensitive pharmacy data and allows authorization checks to be applied before inventory operations are performed.

Performance Benefit

The separation of responsibilities allows frequently used operations such as inventory lookup, FEFO batch selection, and expiry checking to be handled by dedicated components. This avoids unnecessary processing in the user interface and allows the system to efficiently retrieve and update pharmacy information through the appropriate layers.

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